Ex: 7 Flow control at Data link layer using sliding window protocal

Aim:-

To write a program to implement flow control at the data link layer using the sliding window protocol.

Algorithm:-

1) Start the program
2) Read the total number of frames(N), window size(WS), the frame Number to be lost
3) Initialize:-
   base = 0 - first unacknowledged frame
   nextseq = 0 - next frame to be sent
4) while base < N, determine the sender window from base to base + w - 1
5) Send all frames within the current window and increment next seq after sending each frame.
6) If the specified frame is lost
   * Reciver detect the missing frame
   * Reciver discard the out-of-order frames
   * Sender experience a timeout
   * Sender retransmits the frame starting from the lost frame.

Output:

Enter total number of frames to Send: 3
Enter Sender window size : 2
Enter frames number to simulate as lost
(0 to 2, enter -1 for no loss): 1

Sender window : Frame 0 - Frame 1
Node A -> Sending Frame 0
Node B -> Sending Frame 1
(Frame 1 lost in transit)
Node B -> Frame 1 missing
Node A -> Timeout ! Resending frame 1

Sender window : [Frame 1 - Frame 2]
Node A -> Sending Frame 1
Node A -> Sending Frame 2
Node B -> Recived Frame 1, Send Ack!
All 3 frame successfully transmitted from Node A to Node B

=) If three is no frame loss:
* Reciver accepts the frames in order
* Peirves send an Ack for each received frame
* Sender reciver the Ack slides the Window forward.
-> set base = nextseq.

8) Repeat step 4-7 untill all frames are successfully transmitted & acknowleged
9) Display that all frames
10) Stop the program.

Code:-

```
Import java.util.Scanner;
Public class SlidingWindow{
Static void Simulate (int totalframes,
    int windowsize, Integer lostFrame){
    int base=0;
    int nextseq=0;
    While( base < totalframe){
        int windowEnd = Math.min( base+windowsize-1, totalframes-1);
System.out.println("Sender window : [Frame"+base+"-Frame"+windowEnd+"]")
While (nextseq < Base + windowsize && nextseq < totalframes){
System.out.println(" Node A --> Sending frames"+nextseq);
```

```
if (lostFrame != null && nextseq == lostFrame){
    System.out.println("Frame " + nextseq + "Lost in transit");
}
} next seq;

if (lost Frame != null && base <= lost Frame &&
        lost frame < nextseq){
    System.out.println("Node B => Frame" + lost Fram
    + " missing");
    System.out.println("Node A -> Timeout! Resendig
        window Starting from Frame" + lost Fra
        ");
    next seq = lost Frame;
    lost Frame = null;
        continue;
    for (int f = base; f < nextSeq; f++){
        System.out.println("Node B -> Recived Frame" +
    f + ", Sending Ack" + f); }
    base = next seq; );
    } }
Public static void Main (String [] args){
    Scanner sc = new Scanner (System.in);
    System.out.println("Enter total number of frams to
end");
    int total frames = sc.nextInt();
    System.out.println("Enter from number to simulate as
                lost \n" + 0 to " +
        (total frames - 1) + ", enter -1 for no lost;
        );
} else{
    lost frame = input.nextInt(); }
    Simulate (total Frame, window size,
    sc.close(); } }          lost frame);
```

Result:

Thus the program to implement flow control at the Data link layer using the sliding window protocol was implemented succesfully.